

TRENT B. THOMAS

Crosby Distinguished Postdoctoral Fellow at MIT

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RESEARCH INTERESTS

Planetary science • Low-temperature geochemistry • Evolution of planetary atmospheres • Coupled modeling of planetary interiors, surfaces, and atmospheres • Geochemical cycles • Paleoclimate • Complex systems

EDUCATION

UNIVERSITY OF WASHINGTON Seattle, WA
Ph.D., Earth and Space Sciences, Astrobiology (Dual-Title) 2026

UNIVERSITY OF CALIFORNIA, LOS ANGELES Los Angeles, CA
B.S., Astrophysics, *Phi Beta Kappa* 2020

PROFESSIONAL EXPERIENCE

MASSACHUSETTS INSTITUTE OF TECHNOLOGY Cambridge, MA
Crosby Distinguished Postdoctoral Fellow 2026–present
School of Science Dean's Postdoctoral Fellow 2026–present
Department of Earth, Atmospheric, and Planetary Sciences
• Hosted by O. Jagoutz and R. Pierrehumbert

UNIVERSITY OF WASHINGTON Seattle, WA
NSF Graduate Research Fellow 2020–2026
Department of Earth and Space Sciences
• Advised by D. Catling and V. Meadows
• Ph.D. Thesis: The Maintenance and Restoration of Habitable Conditions on Terrestrial Planets: from Snowball Earth to TRAPPIST-1

NASA JET PROPULSION LABORATORY Pasadena, CA
Research Assistant 2018–2023
Exoplanet Science and Discovery Group
• Advised by R. Hu

AWARDS & HONORS

- Crosby Distinguished Postdoctoral Fellowship, MIT EAPS Department 2026
- Dean's Postdoctoral Fellowship, MIT School of Science 2026
- David A. Johnston Award for Research Excellence, U. Washington ESS Department 2025
- Outstanding Student Poster (finalist), Astrobiology Science Conference 2024
- Winglee Endowed Graduate Fellowship, U. Washington ESS Department 2023
- National Science Foundation Graduate Research Fellowship (NSF GRFP) 2020
- Dean's Prize for Excellence in Undergraduate Research, UCLA 2020
- Early Career Collaboration Award, NASA Astrobiology 2019
- Rudnick-Abelmann Scholarship, UCLA Physics & Astronomy Department 2019

PUBLICATIONS

* Indicates publication in preparation.

- [8] ***Thomas, T. B.**, et al. (*Manuscript in preparation*).
Retrieval of super-*Io* L 98-59 b's interior properties from atmospheric observations.
- [7] Gialluca, M. T., V. S. Meadows, A. P. Lincowski, **T. B. Thomas**, P. C. Hinton, D. Brain, and D. Crisp (2026).
Coupled Photochemical–Climate Modeling of Plausible Tenuous Outgassed Atmospheres on the TRAPPIST-1 Planets. *The Planetary Science Journal*. doi.org/10.3847/PSJ/ae6cef

- [6] **Thomas, T. B.**, F. A. Macdonald, and D. C. Catling (2026). Seafloor Weathering Can Explain the Disparate Durations of Snowball Glaciations. *Geology*. doi.org/10.1130/G53722.1
- [5] **Thomas, T. B.**, V. S. Meadows, J. Krissansen-Totton, M. T. Gialluca, N. F. Wogan, and D. C. Catling (2025). Statistical Geochemical Constraints on Present-day Water Outgassing as a Source of Secondary Atmospheres on the TRAPPIST-1 Exoplanets. *The Planetary Science Journal*. doi.org/10.3847/psj/add261
- [4] Adams, D., M. Scheucher, R. Hu, B. L. Ehlmann, **T. B. Thomas**, R. Wordsworth, E. Scheller, R. Lillis, K. Smith, H. Rauer, and Y. Yung (2025). Episodic Warm Climates on Early Mars Primed by Crustal Hydration. *Nature Geoscience*. doi.org/10.1038/s41561-024-01626-8
- [3] **Thomas, T. B.**, and D. C. Catling (2024). Three-stage Formation of Cap Carbonates after Marinoan Snowball Glaciation Consistent with Depositional Timescales and Geochemistry. *Nature Communications*. doi.org/10.1038/s41467-024-51412-8
- [2] **Thomas, T. B.**, R. Hu, and D. Y. Lo (2023). Constraints on the Size and Composition of the Ancient Martian Atmosphere from Coupled CO₂-N₂-Ar Isotopic Evolution Models. *The Planetary Science Journal*. doi.org/10.3847/psj/acb924
- [1] Hu, R., and **T. B. Thomas** (2022). A Nitrogen-Rich Atmosphere on Ancient Mars Consistent with Isotopic Evolution Models. *Nature Geoscience*. doi.org/10.1038/s41561-021-00886-y

PRESENTATIONS

Invited Talks

Earth and Ocean Sciences Department Seminar, *University of Victoria (2025)*.
 Center for Geoastromy Astrophysics Colloquium, *LMU Munich (2025)*.
 National Oceanography Center Department Seminar, *University of Southampton (2025)*.
 Earth Sciences Department Seminar, *Dartmouth College (2025)*.
 Astrobiology Department Seminar, *University of Washington (2024)*.
 Virtual Planetary Laboratory Seminar, *NASA/UW (2023)*.
 High Performance Computing Seminar, *NASA JPL (2022)*.
 Mars Atmosphere Seminar, *Caltech (2020)*.

Contributed

Thomas, T. B., V. S. Meadows, J. Krissansen-Totton, M. Gialluca, N. Wogan, and D. C. Catling. Statistical constraints on volatile outgassing on terrestrial planets: applications to TRAPPIST-1, SO₂-dominated atmospheres, and the cosmic shoreline. *Astrobiology Science Conference, Madison, Wisconsin (2026)*. Poster.

Thomas, T. B., F. A. Macdonald, and D. C. Catling. Seafloor weathering controls the duration of Neoproterozoic Snowball Earth glaciations. *AGU Fall Meeting, New Orleans, Louisiana (2025)*. Poster.

Thomas, T. B., V. S. Meadows, J. Krissansen-Totton, M. Gialluca, N. Wogan, and D. C. Catling. Water outgassing as a source of secondary atmospheres on the TRAPPIST-1 exoplanets. *STScI Atmospheric Escape and Replenishment Workshop, Baltimore, Maryland (2025)*. Talk.

Thomas, T. B., F. A. Macdonald, and D. C. Catling. Seafloor weathering controls the duration of Neoproterozoic Snowball Earth glaciations. *Life and Planet Conference, London, England (2025)*. Poster.

Thomas, T. B., F. A. Macdonald, and D. C. Catling. Long duration of the ~56 Myr Sturtian Snowball Earth event suggests missing link in the geologic carbon cycle. *European Geoscience Union General Assembly, Vienna, Austria (2025)*. Talk.

Thomas, T. B., and D. C. Catling. A New Model for the Formation of Cap Carbonates after Neoproterozoic Glaciations. *Astrobiology Science Conference, Providence, Rhode Island (2024)*. Poster. **Finalist, Student Poster Competition.**

Thomas, T. B., V. S. Meadows, J. Krissansen-Totton, M. Gialluca, N. Wogan, and D. C. Catling. Constraints on water outgassing rates on the TRAPPIST-1 planets from interior modeling. *Extreme Solar Systems V, Christchurch, New Zealand (2024)*. Poster.

Thomas, T. B., and D. C. Catling. Untangling Planetary Processes in the Neoproterozoic with Cap Carbonates

and a Geologic Carbon Cycle Model. *Goldschmidt Conference, Lyon, France (2023)*. Talk.

Thomas, T. B., R. Hu, and D. Y. Lo. Constraints on the Evolution and Ancient Composition of the Martian Atmosphere from Coupled CO₂–N₂–Ar Isotopic Evolution Models. *54th Division for Planetary Science Conference, London, Ontario, Canada (2022)*. Talk.

Thomas, T. B., and D. C. Catling. A Self-Consistent Model for Generating Marinoan Cap Carbonates and Constraining Neoproterozoic Climate. *Astrobiology Science Conference, Atlanta, Georgia (2022)*. Talk.

Thomas, T. B., R. Hu, and D. Y. Lo. Joint Models for the Evolutionary History of Carbon, Nitrogen, and Argon in the Martian Atmosphere. *53rd Lunar and Planetary Science Conference, The Woodlands, Texas (2022)*. Talk.

Thomas, T. B., and R. Hu. A Nitrogen-Rich Atmosphere on Ancient Mars Indicated by Isotopic Evolution. *AGU Fall Meeting, Virtual (2020)*. Talk.

Thomas, T. B., and R. Hu. A Nitrogen-Rich Atmosphere on Ancient Mars Indicated by Isotopic Evolution. *52nd Division for Planetary Science Conference, Virtual (2020)*. Talk.

Thomas, T. B., and R. Hu. A Nitrogen-Rich Atmosphere on Ancient Mars Indicated by Isotopic Evolution. *UCLA Undergraduate Research Week, Virtual (2020)*. Talk.

Thomas, T. B., and R. Hu. Evolutionary History of the Isotopic Composition of Nitrogen in the Martian Atmosphere. *9th International Conference on Mars, Pasadena, California (2019)*. Poster.

TEACHING & SERVICE

Teaching Experience

UNIVERSITY OF WASHINGTON

Seattle, WA

2024

- Teaching Assistant
ESS 101 Introduction to Geology and Societal Impacts (Prof. Terry Swanson)
Taught two weekly sections of 20–30 students, held weekly office hours.

2023

- Course Creator & Teaching Assistant
ESS 203 Earth's Origin and Transformation over 4.6 Billion Years (Prof. Mark Richards)
Created ten 80-minute lectures, syllabus, and other course material.
Performed all teaching assistant duties. Guest lectured “The history of life on Earth.”

COYOTE CENTRAL YOUTH ARTS ORGANIZATION

Seattle, WA

Summer 2024, 2025, 2026

- Course Creator & Instructor
Generative Design: Creating Art with Code
Created and taught a 20-hour course in Python for K–12 students with no prior coding experience.
See final projects: [2024](#), [2025](#).

Service Experience

- Peer reviewer for *Nature Communications*, *Icarus*, and *The Open Journal of Astrophysics*
- Early Career Investigator Committee, NASA LIFE Research Coordination Network 2025–Present
- Expert Screener, CDRXIV, Preprints and Data for Carbon Dioxide Removal 2025–Present
- Primary Convener, “Global Environmental Changes and Increased Biological Complexity in the Neoproterozoic and Paleozoic”, AbSciCon (2024) & AGU Fall Meeting (2025)
- ESS Department Representative, UW College of the Environment Student Advisory Council 2024–2025
- Student Representative on Faculty Search Committee for Department of Earth and Space Sciences 2024
- Contributor, NASA NExSS & NASA NFoLD Science Communication 2022–2023
- Committee Member, UW Earth and Space Sciences 2020–2026
Awards, computing, graduate-nominated speaker (×2), retreat, peer mentor

MENTORSHIP

Undergraduates

- Veronica Fula (University of Washington).
- Jasmine Singh (Purdue University). Now graduate student at UT San Antonio.

COMMUNITY OUTREACH

- Guest Speaker, Everett Rock and Gem Club 2025
- Science Guest, Bandit Theater, Mad Science Improv 2025
- Speaker (“Mars: Why the Hype?”), Astronomy on Tap, Seattle 2022
- Volunteer Teacher (8 hours, 120 students), Nelson Middle School, Seattle 2022
- Page Creator (Prebiotic atmosphere), Wikipedia 2022
- Invited Speaker, Delran School System Family STEM Night 2022
- Social Media Manager, UW Astrobiology 2021–2022
- Creator & Moderator, UW Astrobiology Public Science Panel Series 2021
- Volunteer Guide, UCLA Planetarium 2019–2020
- Volunteer Scientist, UCLA Exploring Your Universe 2019

ADDITIONAL TRAINING

- Mars Analog Workshop, *UW Astrobiology (2023)*.
- Sagan Summer Workshop, *NASA Exoplanet Science Institute (2023)*.
- Origin of Life Workshop, *UW Astrobiology (2022)*.
- Storytelling Fellows Podcasting Workshop, *UW Libraries (2022)*.
- Planetary Exploration Mission Design Workshop, *UW Astrobiology (2022)*.
- VPlanet Developers Workshop, *Virtual Planetary Laboratory (2021)*.
- ROCKE-3D GCM Tutorial, *NASA GISS (2021)*.
- Quantitative Habitability Workshop, *NASA NExSS (2020)*.
- Exoclims Simulation Platform Summer School, *University of Bern (2019)*.

MEDIA COVERAGE

- *GSA News*, R. Molinek: Understanding “Snowball Earth” Extreme Climates 2026
- *UW News*, H. Hickey: Explaining dramatic planetwide changes after world’s last ‘Snowball Earth’ event 2024
- *NASA Astrobiology*, A. Gronstal: The Size and Shape of Mars’ Ancient Atmosphere 2023
- *LPI Planetary News*: Isotopic Evidence that Ancient Mars’ Atmosphere was More Earth-Like 2022